

Business Context

The problems being solved by Bella's Bistro have varying reasons behind them.

Provide customers with information about the restaurant, menu and special offers: The problem being solved here is the current lack of a digital platform for customers to view and understand completely what is available to them like the international dishes and specific restaurant details. Thus providing this information allows customers to make informed dining decisions and stay updated on special events available improving their overall experience. Subsequently, the business maximises its potential return as customers are making use of all resources available and made known to them.

Allow customers to browse and order food for dine-in or delivery: The problem being solved here is the lack of platforms that allow this level of multifunctionality where a customer can browse, order food and choose how it gets to them or how they get to it and at what times. This streamlines the overall experience for both the business and customers as customers can order food for a dine-in meal at a specific time so when they arrive, they do not have to spend long amounts of time going through the menu or waiting for food rather they can focus on the experience knowing the food is ready. The restaurant benefits massively from this as the system would allow them to accommodate larger than expected customers as some order delivery and some book in a specific time to dine in thus ensuring profit maximisation and assisting workers to prepare foods of the highest priority(delivery) first before preparing for those dining in at a later time, also ensuring dine-in customers receive fresh food that has not been sitting for long increasing likelihood of positive user reviews for the business.

Enable customers to reserve tables for dine-in service: The problem being solved here is ensuring that customers who have had busy days and only have a specific free slot in their schedule can reserve a table giving them peace of mind knowing that they will not need to hope that the restaurant is full but rather have a guaranteed table ready for them. This benefits the business by allowing it to accommodate multitudes of individuals and their specific time availabilities increasing customers and income generated.

Allow customers to browse the menu and place orders for delivery or pickup: The problem being solved here is the modern expectations, customers expect a mobile first on demand service so restaurants without this feel outdated or inconvenient. Subsequently, this avoids operational overload as staff will not need to be taking calls during peak periods which could cause mistakes and stress. This ensures that the business in turn misses no potential revenue.

Let customers reserve tables, view availability and receive booking confirmations: The problem being solved here is the inefficiency of manual bookings. By implementing an online reservation system, users have a convenient way to secure a table and view real-time availability, reducing their wait time and increasing the restaurant's operational efficiency also ensuring no mix-up of bookings takes place thus the business keeps a positive and organised public image.

Allow customers to create accounts to save favourite dishes, past orders, and payment methods: The problem being solved here is the lack of personalisation and data storage for returning users, allowing account creation allows customers to save all their favourite dishes and payment methods making future payments easier and more seamless allowing the business to build a strong customer loyalty base.

Offer rewards for frequent customers, such as discounts or free items: The problem being solved here is low customer retention. By rewarding frequent customers with discounts or free items through a digital loyalty scheme, the business can build a dedicated customer base and encourage repeat visits

Allow customers to rate dishes and leave reviews: The problem being solved here is the lack of insight into customer satisfaction thus by providing a rating and review system for dishes, The business can identify popular menu items and areas for improvement based on direct customer data.

Project Aim

The aim of the project is to provide the client with a fully responsive web-based application for Bella's Bistro that essentially modernises their service delivery and streamlines its internal and external operations. To ensure it is functional the solution will have to be able to:

- i) Provide information about the restaurant, menu and special offers to customers.
- ii) Allow the customers to browse these options and order food for either dine-in or delivery.
- iii) Enable customers to reserve tables for the dine-in service.
- iv) Allow online ordering where customers place orders for delivery or pick-up digitally.
- v) Facilitate table reservations and a confirmation acknowledgement email after a successful booking.
- vi) Ensure customers have the ability to create accounts and save their favourite meals and payment history and details simultaneously.

- vii) Have loyalty programs to offer rewards to frequent customers.
- viii) Allow customers to rate dishes and leave reviews.

Functional Requirements

- i) Allow users to create and manage a customer account using an email and password.
- ii) Allow users to view full international menu meals and including prices and the description.
- iii) Allow customers to book a table by selecting a date, time and number of guests attending.
- iv) Add food items to an order and submit the order for takeout or delivery.
- v) Provide a “Search” and “Filter” function to allow customers to sort the menu by food type or specific requirements.
- vi) Send an automated email once a table booking or food order is successfully submitted.
- vii) Allow users to securely save and manage multiple payment methods within their account profile.
- viii) Apply a discount or reward to a customers basket once a specific stated criteria is met.
- ix) Include a special offers section displaying current deals and holiday packages.
- x) Allow customers to rate dishes and leave written reviews after a completed order.

John Roger – New Customer

Josh Victor – Registered (Returning) Customer

Lola Yoga - Kitchen /Floor Staff

Vincent Pierre – Administrator/ Manager

Functional Requirements	User Stories	Justification	User Acceptance Criteria
Allow users to create and manage a customer account using an email and password.	As John Roger I want to create a secure account with my email and a password so that I can save my details for a faster experience next time I visit the site.	By allowing customers to create a secure account, the system can store their information, favourite dishes and payment methods making the checkout process easier in the future thus this solves the problem of lack of	i)The registration page must have an email and password input option. ii) The system must validate the email and password in their correct manner. iii)Upon successful sign up a secure user profile must be

		individuality and personalisation and data storage for users.	created and user must be redirected to their home page or dashboard.
Allow users to view full international menu meals and including prices and the description.	As John Roger I want to view the full international menu and dish descriptions so that I can see what food is available and make an informed decision before ordering or visiting.	Providing a digital international menu allows customers to make more informed decisions before visiting the establishment, it also gives them an idea of the cost of dishes they are interested in so they can adequately prepare, it is eliminating the problem of customers lacking a centralised digital platform to view the restaurants variety of international cuisines.	i)Menu Page displays a list of all available food items. ii)Each items shows price and description. iii)User can easily navigate between categories.
Allow customers to book a table by selecting a date, time and number of guests attending.	As John Roger I want to be able to book a table by selecting a date, time and number of guest attendings so that I can secure a seat at the restaurant conveniently without having to call.	Implementing an online reservation system allows users to secure a table at any time that suits them in a convenient manner without needed to bother restaurant staff thus decreasing staff overload and increasing customer satisfaction. This	i)The booking form must contain input or selection of date, time and guest count. ii)System must prevent users from selecting a date that has already passed.

		completely removes the need of manual booking processes which a tedious and human error is a factor.	
Add food items to an order and submit the order for takeout or delivery.	As John Roger I want to be able to add food items to an order and submit it for takeaway or delivery so that I can enjoy Bella's Bistro food from home with a quick transaction.	By Providing a seamless digital transaction, the business can capture a larger share of the local market that prefers convenience of mobile ordering over phone calls, in turn eliminating all friction in the current takeout and delivery process making life easier for the customers making them more inclined to purchase from a user friendly system like that of Bella's Bistro.	i)User must be able to click an add to basket button. ii)The basket must allow users to takeout, collection or delivery options. iii)Upon submission the order details should be transmitted to the restaurants order management screen immediately.
Provide a "Search" and "Filter" function to allow customers to sort the menu by food type or specific requirements.	As Josh Victor I want a filter or search to allow me to go straight to my favourite dishes or favourite dishes so that I can quickly reorder the usual meals I have without going through a longer process.	Implementing search and filter tools, customers can quickly find their desired dishes making the digital menu much more accessible and user friendly. This solves the problem of difficulty of navigating a large and diverse menu featuring many	i)A search bar must be present at the top of the Menu Page. ii)Users must be able to select filters. iii)The menu display must update immediately to only show items that match the selected search criteria.

		international cuisines for customers.	
Send an automated email once a table booking or food order is successfully submitted.	As Lola Yoga I want to see real time table availability or food orders pending so that I can prepare the dining area and manage the flow of customers and food needed effectively.	A message of acknowledgement ensures that the staff know that this booking has been confirmed and can adequately prepare, likewise this ensures that the customer is 100% certain their booking has been confirmed leaving no room to worry and proof of booking. It solves the problem of false booking claims by customers and lying staff who may have been unprepared for a booked customer thus ensuring no unnecessary conflict arises keeping customer relations positive ensuring a positive public image of the restaurant to potential customers.	i)A confirmation email is sent to customers to confirm booking likewise the restaurant receives one as well with basic details like customer name and booking time or food order.
Allow users to securely save and manage multiple payment methods within their account profile.	As Josh Victor I want to save multiple payment methods to my account profile so that I can checkout instantly using my	Allowing customers to save multiple payment methods securely improves the efficiency of the takeout and delivery process	i)Account must include a payment methods section. ii)Users must be able to add, name and delete card information.

	preferred card without having to re-enter my information.	providing a smoother checkout service encouraging repeat business and eliminated time consuming activities like entering financial details again.	iii) Saved cards must appear as a selectable list during the final checkout stage of an order.
Apply a discount or reward to a customer's basket once a specific stated criteria is met.	As Josh Victor I want to have my rewards and discounts applied to my baskets automatically so that I feel valued as a frequent customer and save money on my orders.	By rewarding frequent customers based on their order history, the business encourages brand loyalty and repeat visits which proves pivotal for long term revenue gain thus solving the problem of low customer retention in such a competitive market.	i) The system must track the number of orders or total spend in the user's account. ii) When a threshold is met, a discount or free item must automatically be added to the basket. iii) The checkout summary must clearly show the reward was applied.
Include a special offers section displaying current deals and holiday packages.	As Vincent Pierre I want to update the special offers section with holiday celebration deals so that I can attract more customers during the peak holiday season times.	By centralising the special events and offers in one location the business can easily market any celebrations or events ensuring customers are aware of best value options available to them removing the difficulty in promoting seasonal deals.	i) Visible Special offers tab ii) The page must display current special offers or available packages, iii) Users must be able to click on preferred option and see more details or add a relevant deal to basket.
Allow customers to rate dishes and leave written	As Vincent Pierre I want to view customer's	This allows for a feedback loop to be available so as	i) Once an order status is marked completed a

reviews after a completed order.	written reviews so that I can see which international dishes are doing well and which are not and improve services using that information.	to compare which dishes are doing well and which are not doing quite well and adapt accordingly, prompting a data-driven menu eliminating the problem of lack of structured feedback regarding food quality.	leave review option pops up. ii)System provides rating scale and a comment box. iii)Submitted reviews are stored on a database and visible to the site administrator.
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Non-Functional Requirements

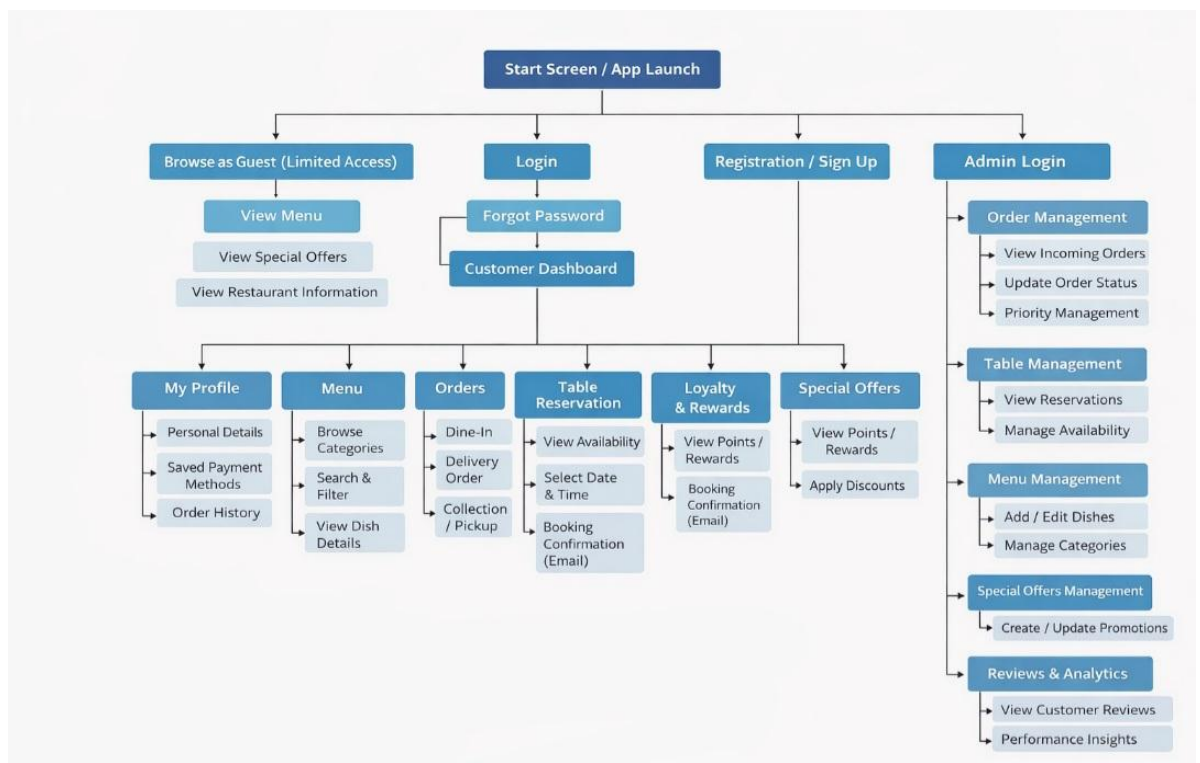
Non-Functional Requirement	Justification	KPI
Security		
System must use HTTPS encryption for all data transmissions.	This is because the system handles sensitive personal data like emails and password, Without encryption hackers could gain access to the data leading to identity theft and a loss of customer trust thus destroying customer relations.	All pages served under HTTPS.
Payment data must be handled using PCI-DSS compliant tokenisation.	Because users can save a plethora of payment methods, the system should not store actual card numbers. Using tokenisation ensures that even in the case of a breach, user data remains secure, protecting the business from legal liability and responsibility.	Zero storage of raw credit card numbers in system.

Performance		
On a 4G+ connection the homepage and menu and corresponding pages should load up within 2 seconds.	Customers in search of delivery and takeout are usually in a rush, If the menu takes too long to load they may simply switch to a faster competitors app causing bad public image and loss of potential customers.	Average page load time less than 2.0 seconds.
System must support atleast 150 users concurrently without a decrease in response time.	During peak times, holidays or celebrations multiple users will be using the website, the system must remain stable and reliable to ensure no orders or tables are lost.	0% latency increase in peak hours.
Reliability		
System should have 99% uptime during Bellas's Bistro operating hours.	If the system goes down during these hours, Operations halt as nothing can be booked and customers cannot place orders. This results in revenue loss and confusion for staff who rely on digital dashboards for orders.	No more than 40 minutes of downtime a month.
System must provide an automatic offline message and redirect to a phone number if the database connection fails.	In the event of a technical failure the business needs a fallback plan to ensure they don't simply lose the customer. Providing a phone number as a backup maintains the professional image and keeps the communication line open.	100% successful trigger of the backup notification during failures.
Accessibility		

System must comply to WCAG 2.1 accessibility standards.	The business serves as a diverse community so without high-contrast ratios and text to speech for users with visual impairments they will not be able to read international menus or book a table. This ensures social inclusion and the business does not face any legal challenges under the equality act.	95% compliance score on accessibility audits.
All interactive buttons must be navigable using only a keyboard.	Many users with mobility impairments cannot use mouse so if the order now or book now buttons are not keyboard-accessible these customers are physically blocked thus loss of potential revenue and business reputation.	Full site navigation possible using only Tab and Enter keys.
Scalability		
The database architecture must be able to support increase from 1000 to 10 000 registered users without structural change.	The project aim is to build long term loyalty. A poorly design database would slow down or crash as it fills data. Scalable architecture ensures the system remains reliable and fast.	Database query response time remains under 100ms as record count increases.
Allow addition of new international menu categories and branches via Admin dashboard.	The business emphasises variety of international dishes. The owners may want to add new dish types. A scalable system allows them to update the menu instantly without needing a developer to change the code.	New menu categories can be added and live within 2 minutes by an admin.
Capacity		

System must provide 50GB+ of storage for HD images of menu options and promotional banners.	To effectively sell international dishes the business needs high-resolution photos to entice customers. Low quality images due to storage limits would make the business look unprofessional.	Storage utilisation remains under 75% capacity at launch.
The system must be capable of storing and retrieving up to 5 years of transaction and review history.	The owners need to look at long term trends to make effective business decisions. If the capacity is too low, valuable data for growth would be lost thus the business using less feedback to improve services.	0% data loss for records up to 60 months old.

Hierarchy Diagram



UI Considerations

Visual Design and Branding

This UI will be used to show the sophisticated nature of Bella's Bistro.

Colours: A warm but luxury colour mix including deep purples, gold accents and clean white backgrounds to give the feeling of a high end dining experience.

Imagery: HD photos of international dishes is a priority, This assists new customers to pick food,

Typography: Easy to read industry standard sans-serif fonts will be used for menu descriptions and prices are clear on both mobile and desktop monitors.

Layout and Navigation

Layout will prioritise a mobile first approach as customer will continuously be on the go when using application.

Navigation: Main menu bar at the top so users can navigate quickly without scrolling up again.

Accessibility

Contrast Ratios: All text to meet WCAG standards for contrast against backgrounds to assist users with visual impairments.

Scalable Text: The UI will allow users to increase font sizes without breaking the page layout.

Alt-Text: Every dish image will include descriptive alt text for screen readers ensuring that visually impaired customers can still experience the menu.

Security Considerations & Risk Mitigation

Data encryption: All data travelling between the customer's browser and Bella's Bistro's server will be protected using a secure SSL/TLS encryption to prevent man in the middle attacks.

Payment Security: No credit card information is stored in local servers rather all payments will be processed through a secure PCI-DSS compliant gateway using secure tokenisation.

Access Control: A role based access control system will be in place so that new customers like John Roger has access to only their profile whilst administrator Vincent Pierre has access to the backend reports and managing the menus.

Secure Authentication: User passwords will be protected using a strong algorithm so that in the event of a data breach the real user passwords are not exposed.

Risk Mitigation Table/ Matrix

Risk Identification	Impact	Probability	Mitigation Strategy
Data Breach: Customer emails and information being leaked	High	Low	Putting in place strict input validation and regular security patching of the web server
Downtime: The site fails during peak hours or holidays.	High	Medium	Using a scalable cloud hosting system with 99% uptime and daily backups in place.
SQL Injection: Users with malicious intent try to manipulate the database.	High	Medium	Use prepared statements and regulated queries for all database interactions to neutralise malicious code.
User Error: Staff accidentally deletes a booking or order item.	Medium	Medium	Implement an Undo feature and a recycle bin system where data is archived and not deleted instantly.
Payment Failure: The payment gateway goes down	Medium	Low	Provide a pay at restaurant or pay on delivery option to ensure the sale and potential revenue is not lost.

Legal and regulatory requirements

Regulation	Application
Data Protection Act(2018)/ GDPR	Users like John Roger's personal data must be protected, thus this includes the right for users to delete their accounts and the requirement that only data necessary for the service like emails or addresses are collected.

Equality Act (2010)	The system must not discriminate. If the User interface is not accessible to those with disabilities the restaurant could be in breach of this act.
Consumer Rights Act	More pertaining to the Takeout and Delivery options, the food described on the digital menu must match the quality of the food delivered, If the description is misleading, the customer has legal grounds to receive a refund.
Food Information Regulations (2014)	Digital menu must clearly flag allergens in descriptions of dishes, This ensures that customers can safely choose international meals they have never tried before without risk to their health.